



IT 123 – System Administration and Maintenance

Laboratory#6

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- Screenshots of netplan config (before and after).

```
GNU nano 8.7.1                                00-installer
# This is the network config written by 'subiquity'
network:
  ethernets:
    enp0s3:
      dhcp4: true
      dhcp6: true
      match:
        macaddress: 08:00:27:4c:3f:fb
      set-name: enp0s3
  version: 2
```

UbuntuServer [Running] - Oracle VirtualBox
File Machine View Input Devices Help

```
robin@UbuntuServer:/etc/netplan$ sudo netplan apply
robin@UbuntuServer:/etc/netplan$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:4c:3f:fb brd ff:ff:ff:ff:ff:ff
    altnames enx000274c3ffb
    inet 192.168.1.150/24 brd 192.168.1.255 scope global enp0s3
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe4c:3ffb/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
robin@UbuntuServer:/etc/netplan$ ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data:
64 bytes from 8.8.8.8: icmp_seq=1 ttl=117 time=12.6 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=117 time=9.09 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=117 time=12.8 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=117 time=12.8 ms

--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 9.894/12.020/12.843/1.231 ms
robin@UbuntuServer:/etc/netplan$ nslookup google.com
Server:      127.0.0.53
Address:     127.0.0.53#53

Non-authoritative answer:
Name:   google.com
Address: 172.217.26.238
Name:   google.com
Address: 2404:6800:4017:80b::200e
robin@UbuntuServer:/etc/netplan$
```



- Ping and traceroute results.

```
robin@UbuntuServer:/etc/netplan$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data:
64 bytes from 8.8.8.8: icmp_seq=1 ttl=117 time=12.1 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=117 time=9.83 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=117 time=11.8 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=117 time=9.90 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=117 time=12.0 ms
64 bytes from 8.8.8.8: icmp_seq=6 ttl=117 time=11.8 ms
64 bytes from 8.8.8.8: icmp_seq=7 ttl=117 time=12.8 ms

64 bytes from 8.8.8.8: icmp_seq=8 ttl=117 time=12.8 ms
64 bytes from 8.8.8.8: icmp_seq=9 ttl=117 time=11.7 ms
64 bytes from 8.8.8.8: icmp_seq=10 ttl=117 time=9.86 ms
64 bytes from 8.8.8.8: icmp_seq=11 ttl=117 time=10.8 ms
64 bytes from 8.8.8.8: icmp_seq=12 ttl=117 time=9.88 ms
64 bytes from 8.8.8.8: icmp_seq=13 ttl=117 time=11.8 ms
^X64 bytes from 8.8.8.8: icmp_seq=14 ttl=117 time=19.7 ms
64 bytes from 8.8.8.8: icmp_seq=15 ttl=117 time=9.45 ms
64 bytes from 8.8.8.8: icmp_seq=16 ttl=117 time=12.9 ms
64 bytes from 8.8.8.8: icmp_seq=17 ttl=117 time=11.7 ms
^C
--- 8.8.8.8 ping statistics ---
17 packets transmitted, 17 received, 0% packet loss, time 16024ms
rtt min/avg/max/mdev = 9.454/11.808/19.661/2.258 ms
robin@UbuntuServer:/etc/netplan$ traceroute 8.8.8.8
Command 'traceroute' not found, but can be installed with:
sudo apt install traceroute # version 1:2.1.6-1build1, or
sudo apt install inetutils-traceroute # version 2:2.1.4-2ubuntu1.1
robin@UbuntuServer:/etc/netplan$
```

- DNS resolution test (**nslookup**).

```
robin@UbuntuServer:/etc/netplan$ nslookup google.com
Server:      127.0.0.53
Address:     127.0.0.53#53

Non-authoritative answer:
Name:   google.com
Address: 172.217.26.78
Name:   google.com
Address: 2404:6800:4017:80b::200e

robin@UbuntuServer:/etc/netplan$
```